Skin Lesion Severity Classifier

Project Overview:

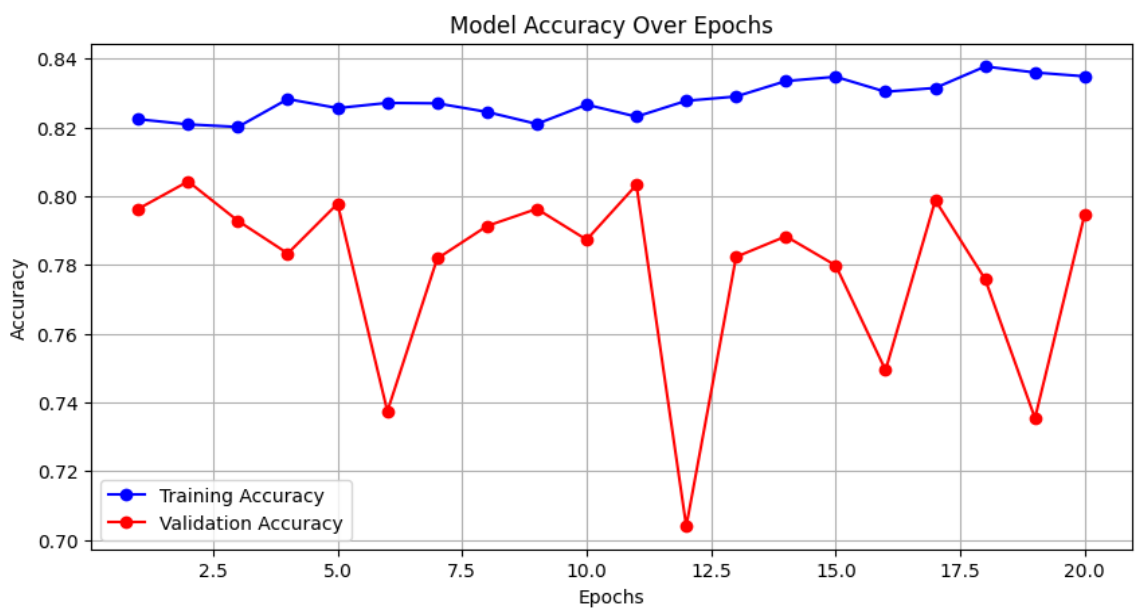
The Skin Lesion Severity Classifier is a medical imaging project that utilizes Deep Learning to assist in the early detection of skin cancer. By analyzing dermoscopy images from the HAM10000 dataset, the system classifies lesions into different diagnostic categories to help prioritize high-risk cases. The primary goal is to provide a preliminary screening tool that can distinguish between harmless marks and potentially malignant conditions through a high-interaction Streamlit dashboard.

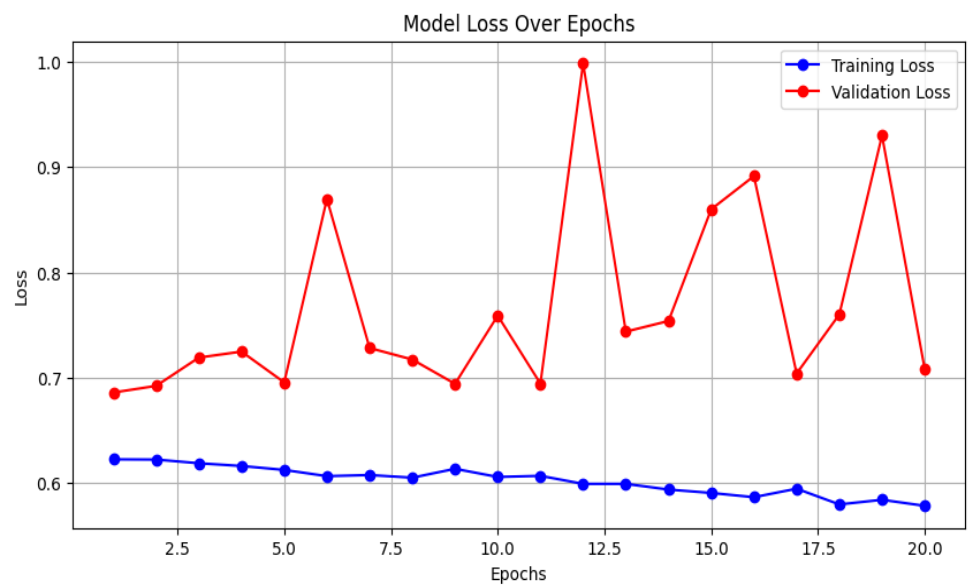
Exploratory Data Analysis (EDA) & Model Setup:

The project began by processing the HAM10000 metadata to understand class distribution and lesion localization.
Data Preparation: Images were resized to 128x128 pixels to maintain high-resolution textures necessary for medical diagnosis.
Architecture: The model features three convolutional blocks with 64, 128, and 256 filters, respectively.
Optimization: Techniques such as Batch Normalization, L2 Regularization, and Dropout were implemented to ensure the model remains stable and generalizes well to new data.

Training Progress and Learning Curves:

The model's performance was tracked over **20 epochs**, focusing on its ability to learn and adapt to the complex patterns in skin lesions.
Accuracy Analysis: According to **ACCURACY.png**, the training accuracy reached a stable peak of approximately **84%**. The validation accuracy fluctuated between **70% and 80.5%**, reflecting the inherent difficulty in classifying visually similar medical conditions.
Loss Analysis: As shown in **LOSS.png**, the training loss decreased steadily from **0.62 to 0.58**, indicating consistent learning. A significant spike in validation loss occurred around **Epoch 12**, identifying a period where the model struggled with specific validation samples before stabilizing again.

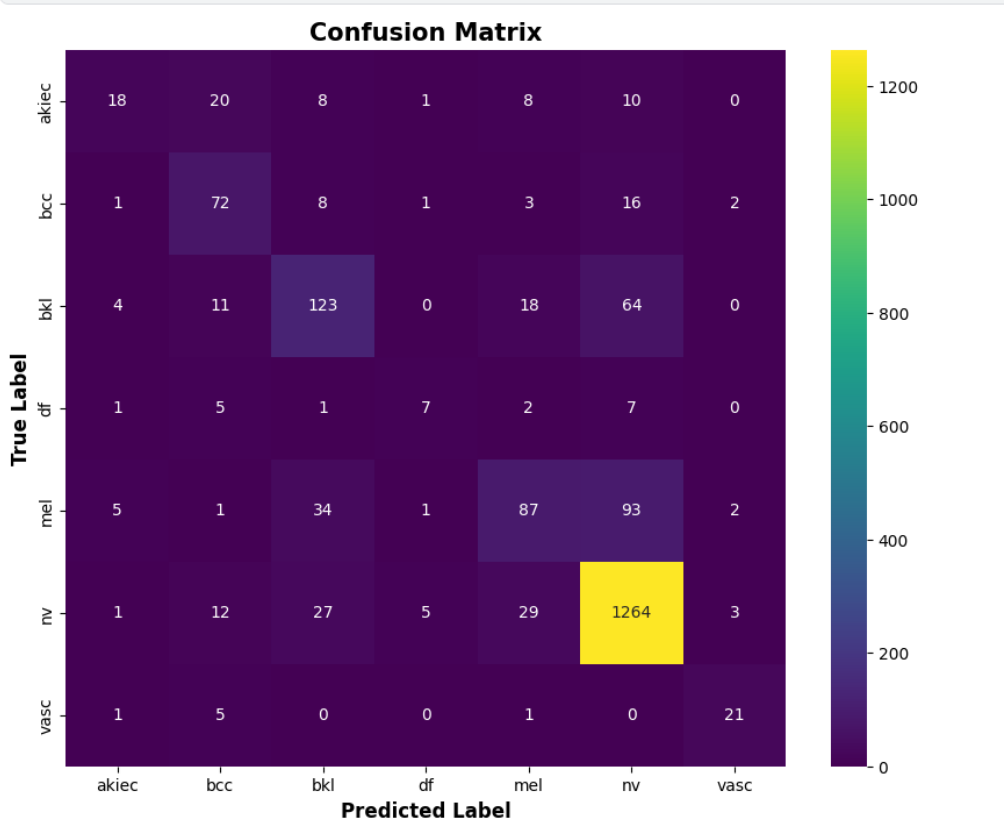




Model Evaluation and Diagnostic Insights:

The evaluation phase utilized a Confusion Matrix to visualize exactly where the model succeeded and where it faced challenges.

- **Top Performing Class:** As seen in **confusion_matrix.png**, the model is exceptionally accurate at identifying **Melanocytic nevi (nv)**, with **1,264** correct predictions.
- **Benign vs. Malignant Overlap:** The model successfully identified **87** cases of **Melanoma (mel)** but misclassified **93** cases as **nv**. This suggests that while the model is strong at detecting benign lesions, there is a shared visual similarity between certain malignant and harmless cases.
- **Other Lesion Types:** The model showed high precision for **Vascular lesions (vasc)** with **21** correct identifications and solid performance for **Basal cell carcinoma (bcc)** with **72** correct cases.



Conclusion:

The **Skin Lesion Severity Classifier** successfully demonstrates the power of Convolutional Neural Networks in medical triaging. With a training accuracy of **84%**, the model serves as a robust prototype for preliminary skin assessment. While the current version excels at identifying benign lesions, future iterations can be improved by using data augmentation to better distinguish between Melanoma and benign nevi. The integration of this model into a **Streamlit** interface makes it a highly accessible portfolio piece that bridges the gap between AI research and practical healthcare applications.